

Patient-Reported Outcomes, Treatment Persistence, and Health-Resource Utilization With Cardionex in Adults With Chronic Heart Failure: A Global Observational Study

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Abstract

Chronic heart failure imposes a persistent burden on daily functioning, medication adherence, healthcare utilization, and patient confidence in long-term disease management. Cardionex, the drug name associated with Corvanta, is a simulated investigational therapeutic concept developed for internal scientific-content workflow demonstration in chronic heart failure management. This simulated global observational study evaluated patient-reported outcomes, treatment persistence, adherence patterns, functional status, care-pathway integration, and health-resource utilization among adults receiving Cardionex as part of structured chronic heart failure management. Adults with chronic heart failure and ongoing symptom burden were enrolled across a simulated global care network and followed for 48 weeks. Outcomes included changes in symptom impact, quality of life, self-reported functional confidence, adherence, persistence, outpatient care escalation, urgent care visits, and heart failure hospitalization. A total of 1,248 simulated participants were included. At week 48, mean symptom impact score improved by –13.4 points, quality-of-life limitation score improved by –11.1 points, and functional confidence increased by 18.6 points. Treatment persistence at week 48 was 78.9%, and high adherence was observed in 72.4% of participants. Heart failure-related urgent care visits occurred in 14.2%, and hospitalization occurred in 9.6%. Findings suggest that Cardionex-based simulated care may support patient-centered stability, persistence, and care coordination. These data are simulated and intended only for internal evidence mapping and medical-affairs workflow demonstration.

Keywords: Chronic heart failure; patient-reported outcomes; Cardionex; Corvanta; treatment persistence; adherence; real-world evidence; health-resource utilization

Introduction

Chronic heart failure is a long-term clinical syndrome that affects physical capacity, emotional well-being, healthcare engagement, medication-taking behavior, and day-to-day quality of life. For cardiologists and

internal medicine specialists, management often extends beyond acute stabilization and pharmacologic optimization. It includes sustained monitoring, patient education, adherence support, recognition of worsening congestion, comorbidity management, and coordination across outpatient and urgent-care settings.

Traditional clinical endpoints such as hospitalization and mortality remain important, but patient-reported outcomes provide essential insight into how chronic heart failure affects daily life. Patients frequently describe fatigue, shortness of breath, reduced confidence in physical activity, sleep disruption, anxiety about disease worsening, and difficulty maintaining complex care plans. These experiences can influence adherence, persistence, care engagement, and healthcare-resource utilization.

Real-world evidence and observational study designs are especially relevant for chronic heart failure because patient behavior, comorbidity burden, background therapy, social support, and care-access patterns can meaningfully shape outcomes. In routine practice, treatment benefit is not determined only by clinical response; it is also influenced by whether patients can remain on therapy, tolerate monitoring requirements, follow care instructions, and avoid repeated urgent-care escalation.

Corvanta is the brand name associated with Cardionex, a simulated investigational therapeutic concept created for internal scientific-content workflow demonstration. Cardionex is positioned as a chronic heart failure management concept intended to support patient-centered stability, adherence, persistence, and care-pathway coordination when evaluated within a structured chronic-care framework.

The objective of this simulated global observational study was to evaluate patient-reported outcomes, treatment persistence, adherence patterns, functional status, care-pathway integration, and health-resource utilization among adults receiving Cardionex as part of chronic heart failure management.

Methods

Study Design

This was a simulated, global, prospective observational study conducted over 48 weeks. The study evaluated adults with chronic heart failure who initiated Cardionex within a structured care model. The observational design was selected to reflect patient-centered outcomes, adherence behavior, persistence, care interactions, and health-resource utilization in a simulated routine-practice environment.

This manuscript contains simulated data generated for internal scientific-content workflow demonstration. The study design, population, endpoints, and results are hypothetical and should not be interpreted as real-world clinical evidence, product efficacy, product safety, regulatory support, guideline endorsement, or prescribing guidance for Corvanta or Cardionex.

Study Population

The simulated population included adults with chronic heart failure who had ongoing symptom burden and were being managed by cardiologists, internal medicine specialists, or integrated chronic-care teams. The population was designed to reflect a broad global chronic heart failure care setting, including patients with

variable functional limitation, prior hospitalization history, comorbidities, and background therapy complexity.

Inclusion Criteria

Participants were eligible for inclusion if they met all of the following simulated criteria:

- Adults aged 40 years or older.
- Documented chronic heart failure for at least 6 months.
- Persistent patient-reported symptom burden despite background care.
- Initiation of Cardionex within a structured simulated chronic heart failure management pathway.
- Availability of baseline patient-reported outcome assessments.
- Ability to complete follow-up assessments through digital, clinic-based, or telephone-supported methods.
- At least one documented outpatient heart failure care interaction during the baseline assessment period.

Exclusion Criteria

Participants were excluded if they met any of the following simulated criteria:

- Acute decompensated heart failure requiring hospitalization at the time of enrollment.
- Advanced heart failure intervention planned within the first 12 weeks of follow-up.
- Major cardiovascular procedure within 60 days before enrollment.
- Severe cognitive or communication limitation preventing patient-reported outcome assessment.
- Unstable non-cardiac illness expected to dominate clinical outcomes.
- Incomplete baseline care-pathway documentation.
- Participation in another simulated interventional research protocol during the observation period.

Intervention or Exposure Concept

Participants were observed after initiation of Cardionex as part of a simulated structured chronic heart failure management pathway. The care model included baseline symptom assessment, medication-support counseling, renal and electrolyte monitoring, blood pressure review, congestion-status education, adherence support, and scheduled follow-up interactions.

Cardionex exposure in this manuscript is a simulated investigational concept. No dosing, prescribing, or product-use instruction should be inferred from this document.

Endpoints

The primary endpoint was change from baseline to week 48 in the Heart Failure Daily Impact Score, a simulated 0–100 patient-reported scale where higher scores indicate greater disease impact.

Secondary endpoints included:

- Change from baseline in Heart Failure Quality-of-Life Limitation Score.

- Change from baseline in Functional Confidence Index.
- Treatment persistence at weeks 12, 24, 36, and 48.
- Adherence category through week 48.
- Frequency of outpatient care escalation.
- Frequency of urgent care visits for worsening heart failure symptoms.
- Frequency of heart failure hospitalization.
- Patient-reported confidence in self-monitoring and congestion recognition.
- Frequency and type of adverse events reported during follow-up.

Statistical Analysis

The simulated analysis included all enrolled participants with baseline data and at least one follow-up assessment. Continuous outcomes were summarized using means, standard deviations, and mean changes from baseline. Categorical outcomes were summarized using counts and percentages. Persistence was defined as remaining on Cardionex without permanent discontinuation at each follow-up time point. High adherence was defined as simulated adherence of at least 80% during the observation period.

Exploratory subgroup summaries were conducted by baseline symptom burden, age group, prior hospitalization history, baseline functional limitation, and comorbidity burden. Missing patient-reported outcome data were handled using simulated mixed-model repeated-measure summaries and sensitivity analyses based on observed cases.

Ethics and Compliance Statement

This simulated observational study was reviewed under a fictional internal scientific-governance process for content workflow demonstration. No real patients, investigators, clinical sites, ethics committees, registries, regulatory authorities, or real-world health systems were involved. All data and analyses are fictional and were created solely for internal demonstration of medical-affairs evidence generation and content workflow testing.

Results

Participant Disposition and Follow-Up

A total of 1,248 simulated participants were included in the observational cohort. Week 48 follow-up data were available for 1,108 participants. The main reasons for incomplete follow-up were loss to follow-up, participant withdrawal, care transfer, and incomplete patient-reported outcome submission.

Disposition Category	Simulated Participants (N = 1,248)
Enrolled in observational cohort	1,248
Completed week 12 follow-up	1,198 (96.0%)
Completed week 24 follow-up	1,161 (93.0%)
Completed week 36 follow-up	1,132 (90.7%)

Disposition Category	Simulated Participants (N = 1,248)
Completed week 48 follow-up	1,108 (88.8%)
Discontinued Cardionex before week 48	263 (21.1%)
Lost to follow-up	52 (4.2%)
Transferred care outside simulated network	39 (3.1%)
Withdrew from observational follow-up	31 (2.5%)
Incomplete patient-reported outcome submission	58 (4.6%)

Baseline Characteristics

The simulated cohort included adults with chronic heart failure and clinically relevant comorbidity burden. Most participants had at least one major comorbidity, and approximately one-third had a prior heart failure hospitalization within 12 months before enrollment.

Baseline Characteristic	Overall Cohort (N = 1,248)
Mean age, years	67.4 ± 9.8
Age ≥75 years	338 (27.1%)
Female participants	526 (42.1%)
Mean body mass index, kg/m ²	29.7 ± 5.4
Prior heart failure hospitalization within 12 months	407 (32.6%)
Hypertension	918 (73.6%)
Type 2 diabetes	502 (40.2%)
Chronic kidney disease history	383 (30.7%)
Atrial fibrillation history	361 (28.9%)
High comorbidity burden category	429 (34.4%)
Baseline Heart Failure Daily Impact Score	62.8 ± 13.5
Baseline Quality-of-Life Limitation Score	59.6 ± 14.1
Baseline Functional Confidence Index	46.2 ± 16.8

Patient-Reported Outcomes

At week 48, the mean Heart Failure Daily Impact Score improved by –13.4 points from baseline. Improvement was observed by week 12 and appeared sustained through week 48. Quality-of-life limitation improved by –11.1 points, and the Functional Confidence Index increased by 18.6 points.

Patient-reported domains with the most consistent improvement included confidence during light activity, perceived ability to complete daily tasks, reduced worry about symptom worsening, and improved recognition of congestion-related warning signs.

Patient-Reported Outcome	Baseline	Week 24	Week 48	Mean Change at Week 48
Heart Failure Daily Impact Score	62.8 ± 13.5	52.1 ± 14.2	49.4 ± 14.8	-13.4
Quality-of-Life Limitation Score	59.6 ± 14.1	51.3 ± 14.6	48.5 ± 15.0	-11.1
Functional Confidence Index	46.2 ± 16.8	59.7 ± 17.2	64.8 ± 17.6	+18.6
Self-Monitoring Confidence Score	44.9 ± 18.1	60.5 ± 17.4	66.7 ± 16.9	+21.8
Congestion Recognition Confidence Score	41.7 ± 17.5	58.8 ± 17.8	63.9 ± 17.1	+22.2

Treatment Persistence and Adherence

Treatment persistence was 91.8% at week 12, 86.2% at week 24, 82.1% at week 36, and 78.9% at week 48. High adherence through week 48 was observed in 72.4% of participants. The most common reasons for discontinuation were patient preference, adverse events, perceived treatment complexity, care-plan change, and clinical deterioration requiring individualized reassessment.

Persistence and Adherence Outcome	Simulated Participants (N = 1,248)
Persistent on Cardionex at week 12	1,146 (91.8%)
Persistent on Cardionex at week 24	1,076 (86.2%)
Persistent on Cardionex at week 36	1,025 (82.1%)
Persistent on Cardionex at week 48	985 (78.9%)
High adherence through week 48	904 (72.4%)
Moderate adherence through week 48	227 (18.2%)
Low adherence through week 48	117 (9.4%)
Discontinuation due to patient preference	76 (6.1%)
Discontinuation due to adverse event	58 (4.6%)
Discontinuation due to perceived treatment complexity	43 (3.4%)
Discontinuation due to care-plan change	39 (3.1%)

Persistence and Adherence Outcome	Simulated Participants (N = 1,248)
Discontinuation due to clinical deterioration	32 (2.6%)

Care-Pathway Integration

Care-pathway integration was assessed through completion of scheduled monitoring, documentation of symptom review, medication-support interactions, and patient education reinforcement. Most participants completed at least three structured follow-up interactions during the observation period.

Participants with high care-pathway engagement showed greater improvement in patient-reported outcomes and fewer urgent-care events compared with participants with low engagement. These findings are simulated and should be interpreted as internal workflow evidence only.

Care-Pathway Measure	Overall Cohort (N = 1,248)
Completed at least three scheduled follow-up interactions	1,039 (83.3%)
Documented symptom review at each scheduled visit	922 (73.9%)
Completed renal and electrolyte monitoring schedule	986 (79.0%)
Received medication-support counseling	1,084 (86.9%)
Received congestion-status education	1,102 (88.3%)
Used digital or telephone-supported follow-up	776 (62.2%)
Required additional adherence-support contact	318 (25.5%)
Required intensified care coordination	241 (19.3%)

Health-Resource Utilization

During the 48-week follow-up period, heart failure-related urgent care visits occurred in 177 participants, and heart failure hospitalization occurred in 120 participants. Outpatient care escalation for worsening symptoms or congestion concerns occurred in 267 participants. Participants with high adherence and high care-pathway engagement had numerically lower rates of urgent care visits and hospitalization in exploratory simulated summaries.

Health-Resource Utilization Outcome Through Week 48	Simulated Participants (N = 1,248)
Outpatient care escalation for worsening symptoms	267 (21.4%)
Urgent care visit for heart failure symptoms	177 (14.2%)
At least one heart failure hospitalization	120 (9.6%)
More than one heart failure hospitalization	31 (2.5%)

Health-Resource Utilization Outcome Through Week 48	Simulated Participants (N = 1,248)
Emergency assessment without hospitalization	96 (7.7%)
Diuretic or background-care adjustment due to congestion concern	289 (23.2%)
Care-team contact for weight gain or fluid-retention symptoms	342 (27.4%)
Unscheduled cardiology review	196 (15.7%)

Exploratory Engagement-Based Outcomes

Participants with high adherence and high care-pathway engagement showed greater improvement in patient-reported outcomes than participants with low adherence or low engagement. This simulated analysis supports the importance of chronic-care structure, patient education, and follow-up continuity when evaluating long-term heart failure management concepts.

Outcome at Week 48	High Adherence / High Engagement (n = 691)	Low Adherence / Low Engagement (n = 146)
Mean change in Daily Impact Score	-16.2 ± 13.9	-6.1 ± 14.7
Mean change in Quality-of-Life Limitation Score	-13.6 ± 14.2	-4.8 ± 15.1
Mean change in Functional Confidence Index	+22.7 ± 17.3	+7.9 ± 18.6
Urgent care visit for heart failure symptoms	68 (9.8%)	34 (23.3%)
At least one heart failure hospitalization	47 (6.8%)	24 (16.4%)
Outpatient care escalation	112 (16.2%)	49 (33.6%)

Safety and Tolerability

Treatment-emergent adverse events were reported in 612 participants. Most events were mild or moderate. The most frequently reported events were dizziness, fatigue, mild gastrointestinal symptoms, renal-function monitoring alerts, symptomatic hypotension, and electrolyte monitoring alerts. Serious adverse events were reported in 132 participants and were primarily related to heart failure worsening, infection, arrhythmia, renal instability, or other comorbidity-related events.

Safety Outcome Through Week 48	Simulated Participants (N = 1,248)
Any treatment-emergent adverse event	612 (49.0%)

Safety Outcome Through Week 48	Simulated Participants (N = 1,248)
Mild adverse event	311 (24.9%)
Moderate adverse event	236 (18.9%)
Severe adverse event	65 (5.2%)
Serious adverse event	132 (10.6%)
Adverse event leading to discontinuation	58 (4.6%)
Dizziness	119 (9.5%)
Fatigue	103 (8.3%)
Mild gastrointestinal symptoms	97 (7.8%)
Renal-function monitoring alert	86 (6.9%)
Symptomatic hypotension	54 (4.3%)
Electrolyte monitoring alert	49 (3.9%)

Discussion

This simulated global observational study evaluated patient-reported outcomes, treatment persistence, adherence, care-pathway integration, and health-resource utilization among adults receiving Cardionex within a structured chronic heart failure management framework. The findings suggest that Cardionex-based simulated care was associated with improvements in patient-reported daily impact, quality-of-life limitation, functional confidence, self-monitoring confidence, and congestion-recognition confidence over 48 weeks.

Patient-reported outcomes are particularly important in chronic heart failure because they reflect the lived experience of disease burden. Symptoms such as fatigue, dyspnea, reduced endurance, and anxiety about clinical worsening can shape treatment satisfaction, adherence, care engagement, and willingness to continue long-term management. In this simulated analysis, improvements in daily impact and quality-of-life limitation suggest a potential internal narrative around patient-centered disease-management support.

Treatment persistence and adherence are also important because chronic heart failure care often requires multiple medications, monitoring schedules, lifestyle changes, and frequent clinical review. The week 48 persistence rate of 78.9% and high-adherence rate of 72.4% suggest that most simulated participants remained engaged with Cardionex-based care. However, discontinuation due to patient preference, adverse events, perceived treatment complexity, and care-plan change highlights the importance of practical support, shared decision-making, and individualized monitoring.

The care-pathway integration findings are especially relevant for cardiologists and internal medicine specialists. Participants who completed scheduled follow-up, monitoring, symptom review, medication-support counseling, and congestion-status education appeared to have better patient-reported outcomes and fewer urgent-care events in exploratory summaries. This pattern supports the concept that chronic

heart failure management depends not only on therapeutic exposure but also on structured care delivery and patient engagement.

Health-resource utilization outcomes should be interpreted cautiously. Heart failure-related urgent care visits and hospitalization are influenced by disease severity, comorbidities, background therapy, access to care, monitoring practices, and clinician decision-making. The simulated rates reported in this study are intended for internal workflow demonstration only and should not be interpreted as real evidence of resource-use reduction.

Safety findings were consistent with the complexity of a chronic heart failure population. Dizziness, fatigue, mild gastrointestinal symptoms, renal-function monitoring alerts, symptomatic hypotension, and electrolyte monitoring alerts reinforce the need for structured monitoring and individualized care review. These findings may help support balanced internal content that includes both potential patient-centered benefits and appropriate tolerability considerations.

Overall, this simulated observational study provides a useful structure for medical-affairs content development, particularly when generating materials focused on patient experience, adherence support, chronic-care integration, care-team communication, and evidence-based content workflow demonstration.

Conclusion

In this simulated global observational study, adults receiving Cardionex within a structured chronic heart failure management pathway demonstrated improvements in patient-reported daily impact, quality-of-life limitation, functional confidence, self-monitoring confidence, and congestion-recognition confidence over 48 weeks. Treatment persistence and adherence were maintained in a majority of participants, and care-pathway engagement appeared to align with better patient-reported and health-resource utilization outcomes.

These findings are simulated and should be used only for internal scientific-content workflow demonstration, evidence mapping, medical-affairs narrative development, and care-pathway content testing. They should not be interpreted as real-world clinical evidence, confirmed product benefit, regulatory support, guideline endorsement, or prescribing guidance for Corvanta or Cardionex.

Medical Affairs Interpretation

This simulated research paper can support internal medical-affairs workflows by providing a patient-centered evidence asset focused on chronic heart failure management beyond traditional clinical endpoints. It may be useful for developing content that addresses daily disease impact, treatment persistence, adherence support, monitoring, and care coordination.

Potential internal applications include:

- **Patient-experience evidence mapping:** The patient-reported outcome tables can support mapping of symptom impact, quality-of-life limitation, and functional confidence themes.
- **Adherence and persistence messaging:** Persistence and adherence data can help structure balanced internal narratives about long-term chronic-care engagement.

- **Care-pathway content development:** The care-integration section can support materials focused on monitoring, symptom review, medication-support counseling, and congestion education.
- **Health-resource utilization analysis:** Utilization outcomes can be used to test evidence-linking workflows for urgent-care visits, hospitalization, and outpatient escalation.
- **MLR pre-screen demonstration:** The paper can support claim-source mapping and review of whether generated content avoids overstated real-world effectiveness claims.
- **Omnichannel content planning:** Findings may inform HCP emails, patient-support discussion guides, field medical summaries, congress posters, and digital chronic-care education assets.

Any downstream content should clearly preserve the simulated nature of the evidence and avoid implying confirmed clinical benefit or regulatory approval.

Limitations

This manuscript has several important limitations. First, all data are simulated and do not represent real clinical outcomes for Corvanta or Cardionex. Second, the observational design is hypothetical and cannot establish causal effects. Third, patient-reported outcomes, adherence, persistence, and care-pathway engagement were generated for internal workflow demonstration and may not reflect real-world behavior. Fourth, health-resource utilization outcomes are influenced by care access, local clinical practice, patient comorbidities, background therapy, and monitoring intensity, none of which are fully represented in this simulated dataset. Fifth, safety findings are hypothetical and should not be used to infer the actual tolerability profile of any real product.

Because of these limitations, the manuscript should be used only as a controlled internal scientific-content asset for workflow testing, evidence structuring, medical-affairs planning, and content-generation demonstration.

Data Transparency Statement

All study data, participant characteristics, patient-reported outcomes, adherence patterns, persistence rates, health-resource utilization events, safety findings, author names, affiliations, governance statements, and references in this manuscript are simulated for internal scientific-content workflow demonstration. No real patients, investigators, clinical sites, ethics committees, regulatory authorities, trial identifiers, journal records, DOIs, PMIDs, or real-world datasets are associated with this manuscript. The content should not be interpreted as real-world clinical evidence, regulatory support, guideline endorsement, or prescribing guidance for Corvanta or Cardionex.

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